Metabolomics in clinical biology: promises and challenges



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1. PRINCIPLE

Genome
what can happen

→

Transcriptome
What seems to be happening

→

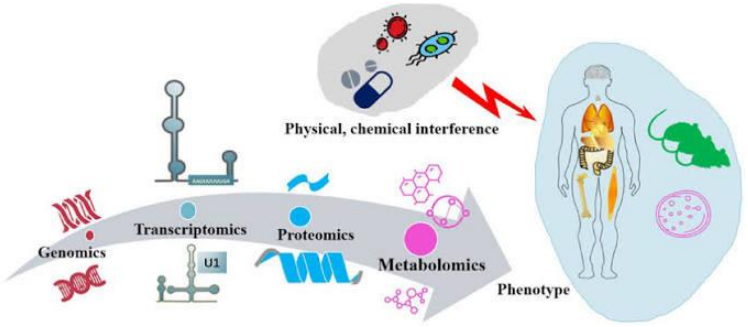
Proteome
What makes things happen

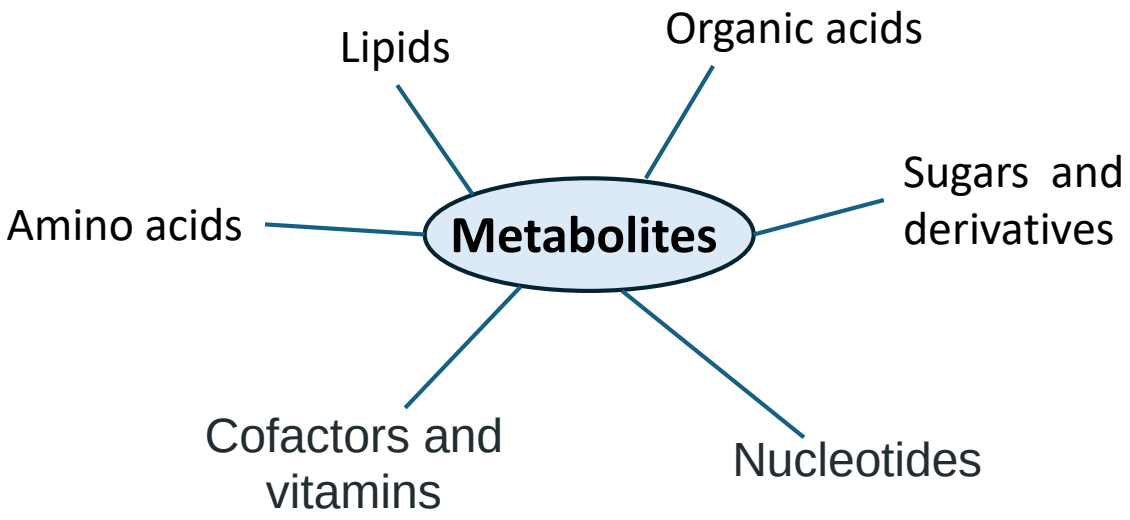
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Metabolome
what really happens

Definition

Qualitative and quantitative analysis of all the small molecules (about > 1,500 Da) in a biological sample.





Exogenous compounds: drugs, toxic (e.g., N_2O)

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what can happen

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Transcriptome
What seems to be happening

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Proteome
What makes things happen

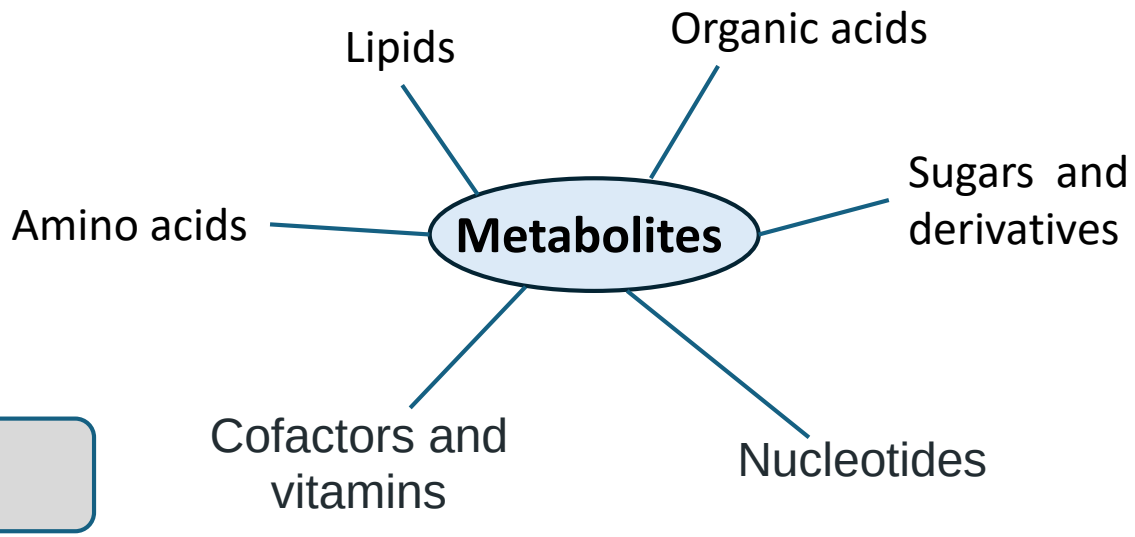
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what really happens

Definition

Qualitative and quantitative analysis of all the small molecules (about > 1,500 Da) in a biological sample.

The role of metabolomics in clinical practice ?



Exogenous compounds: drugs, toxic (e.g., N_2O)



1. PRINCIPLE

From genomics to metabolomics: following the NGS path?

	Research	Standardization	Clinical routine
GENOMICS NGS: exome, genome	• Human genome sequenced (2003) • High-throughput sequencing: cost and time collapse	• Bioinformatic pipelines • ACMG/AMP variant classification, EQA programs	• Rare diseases, oncology panels • France: Plan France Médecine Génomique 2025
METABOLOMICS Targeted and untargeted	• Thousands of features per sample • Many exploratory, single-center studies	• Ongoing: MSI identification levels, QC consortia (mQACC) • Pre-analytical steps still poorly harmonized	• Limited to targeted panels • Amino acids, acylcarnitines (newborn screening), HMM

Metabolomics is on the same path as NGS: standardization is necessary to routine use

STEPS OF A METABOLOMIC ANALYSIS

1

Sampling

Plasma, serum, urine, CSF



2

Preparation

Internal standards, protein precipitation, extraction



3

Separation

Liquid (LC) or gas (GC) chromatography



4

Detection

Mass spectrometry (MS) or nuclear magnetic resonance (NMR)



5

Data

Signal processing, statistics, biological interpretation



Each step adds variability: sampling and preparation must be as standardized as the analysis itself.

1. PRINCIPLE

DISCLAMER : routine metabolomics is not biomarker discovery

BIOMARKER DISCOVERY

Research

- Hypothesis-generating, mostly untargeted
- Relative quantification between groups
- Multivariate statistics, false discovery control
- Output: candidate biomarkers at group level
- Requires validation in independent cohorts

≠

ROUTINE METABOLOMICS

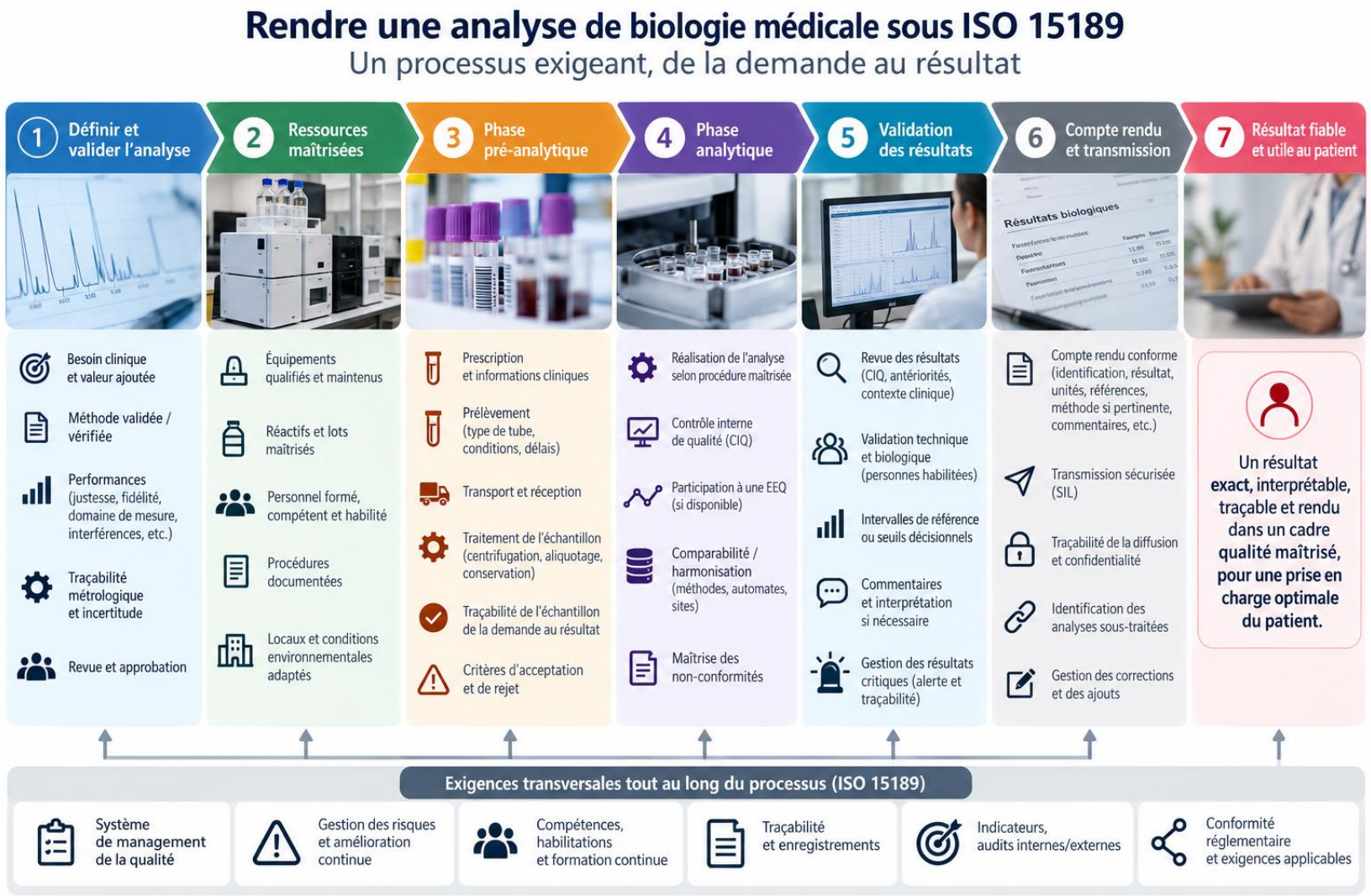
Clinical laboratory

- Targeted, validated method (ISO 15189)
- Absolute quantification: calibrators, labeled internal standards
- Reference intervals, internal QC and external quality assessment

ISO 15189

> A discovery signal becomes a routine biomarker only after targeted validation

1. PRINCIPLE


Rendre une analyse de biologie médicale sous ISO 15189
Un processus exigeant, de la demande au résultat
1 Définir et valider l'analyse
2 Ressources maîtrisées
3 Phase pré-analytique
4 Phase analytique
5 Validation des résultats
6 Compte rendu et transmission
7 Résultat fiable et utile au patient
Résultats biologiques
Besoin clinique et valeur ajoutée
Méthode validée / vérifiée
Performances (justesse, fidélité, domaine de mesure, interférences, etc.)
Traçabilité métrologique et incertitude
Revue et approbation
Équipements qualifiés et maintenus
Réactifs et lots maîtrisés
Personnel formé, compétent et habilité
Procédures documentées
Locaux et conditions environnementales adaptés
Prescription et informations cliniques
Prélèvement (type de tube, conditions, délais)
Transport et réception
Traitement de l'échantillon (centrifugation, aliquotage, conservation)
Traçabilité de l'échantillon de la demande au résultat
Critères d'acceptation et de rejet
Réalisation de l'analyse selon procédure maîtrisée
Contrôle interne de qualité (CIQ)
Participation à une EEQ (si disponible)
Comparabilité / harmonisation (méthodes, automates, sites)
Maîtrise des non-conformités
Revue des résultats (CIQ, antériorités, contexte clinique)
Validation technique et biologique (personnes habilitées)
Intervalles de référence ou seuils décisionnels
Commentaires et interprétation si nécessaire
Gestion des résultats critiques (alerte et traçabilité)
Compte rendu conforme (identification, résultat, unités, références, méthode si pertinente, commentaires, etc.)
Transmission sécurisée (SIL)
Traçabilité de la diffusion et confidentialité
Identification des analyses sous-traitées
Gestion des corrections et des ajouts
Un résultat exact, interprétable, traçable et rendu dans un cadre qualité maîtrisé, pour une prise en charge optimale du patient.
Exigences transversales tout au long du processus (ISO 15189)
Système de management de la qualité
Gestion des risques et amélioration continue
Compétences, habilitations et formation continue
Traçabilité et enregistrements
Indicateurs, audits internes/externes
Conformité réglementaire et exigences applicables



TARGETED OR UNTARGETED: two different approaches

Criterion	TARGETED	UNTARGETED
Approach	***A priori* hypothesis:** we know what we are looking for	**Not *a priori* hypothesis**, hypothesis-generating
Coverage	A few to several hundred defined metabolites	Hundreds to thousands of signals, including unknowns
Quantification	**Absolute** (µmol/L): labeled internal standards and calibration curves	**Relative:** intensities compared between groups
Instrumentation	Triple quadrupole or QTRAP, **MRM mode**	High resolution (**QTOF, Orbitrap**), full-scan analysis
Purpose	Diagnosis, monitoring, biomarker validation	Research, biomarker discovery, phenotyping

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The distinction in metabolite coverage between targeted and untargeted metabolomics is not always clear-cut

Targeted vs untargeted: advantages and limits

	TARGETED	UNTARGETED
+ Advantages	• Absolute quantification (µmol/L) • High sensitivity and specificity (MRM) • Robust, validatable, routine-ready • Direct clinical interpretation	• Broad coverage, hypothesis-free • Discovery of unexpected pathways • Retrospective reanalysis of the data • Phenotyping of the whole metabolic state
− Limits	• Only known, preselected metabolites • A standard needed for each analyte • No discovery beyond the panel	• Relative quantification only • Annotation bottleneck: many unknown features • Batch effects, complex processing, false discoveries • Findings need targeted validation

1. APPLICATION

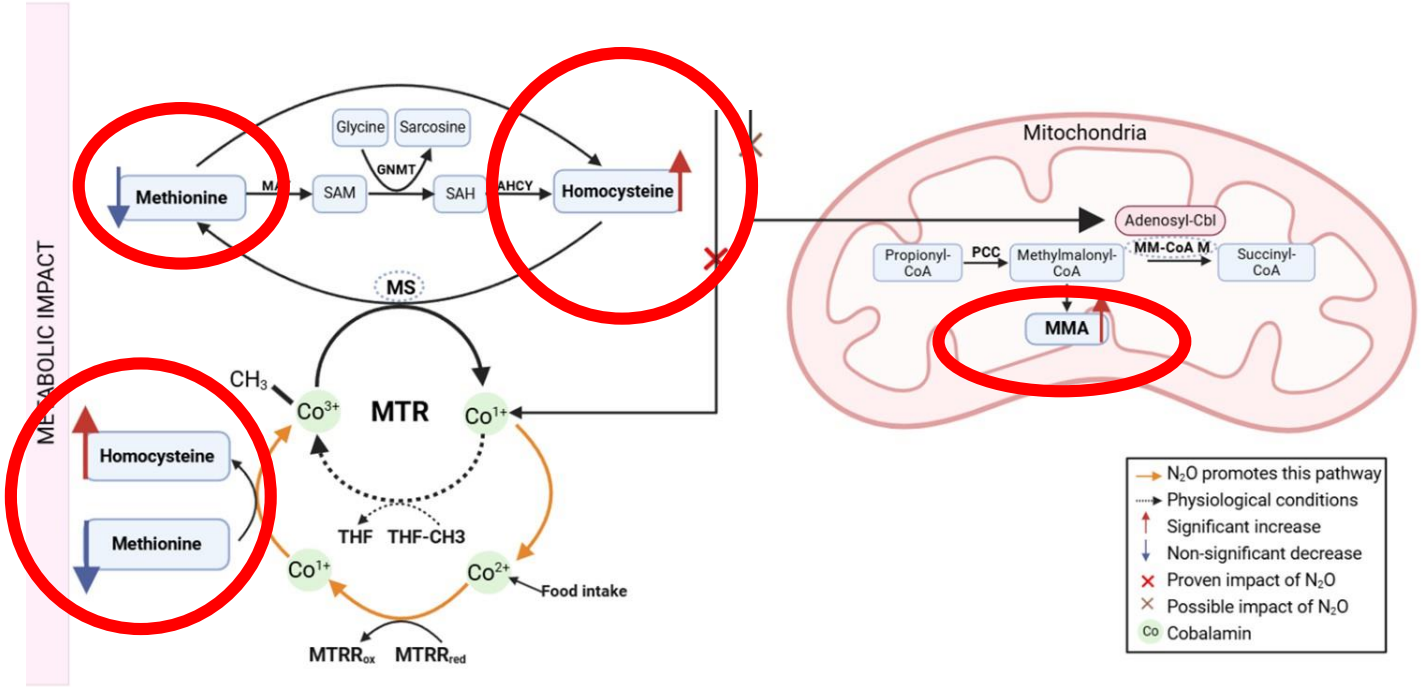
The HMM panel



H
Homocysteine
Substrate of methionine synthase

M
Methionine
Product of remethylation, precursor of SAM

M
Methylmalonic acid
Marker of methylmalonyl-CoA mutase dysfunction



The HMM panel - interpretation

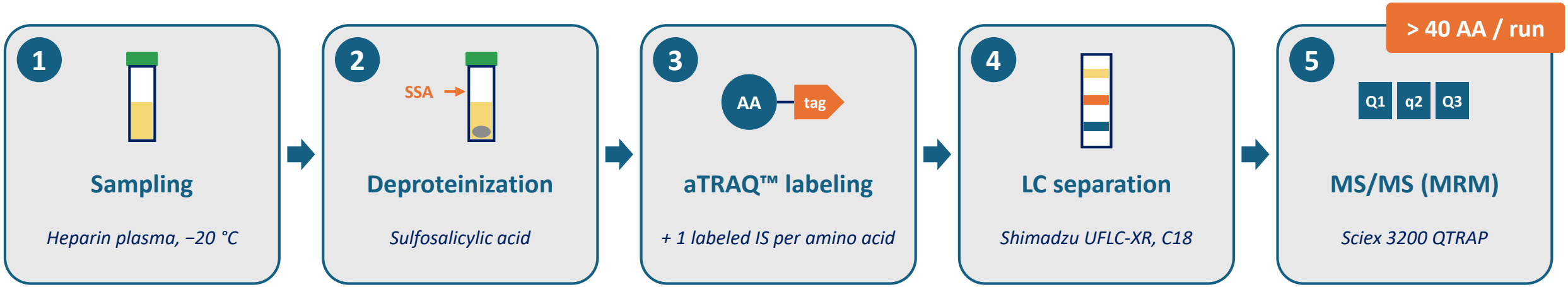
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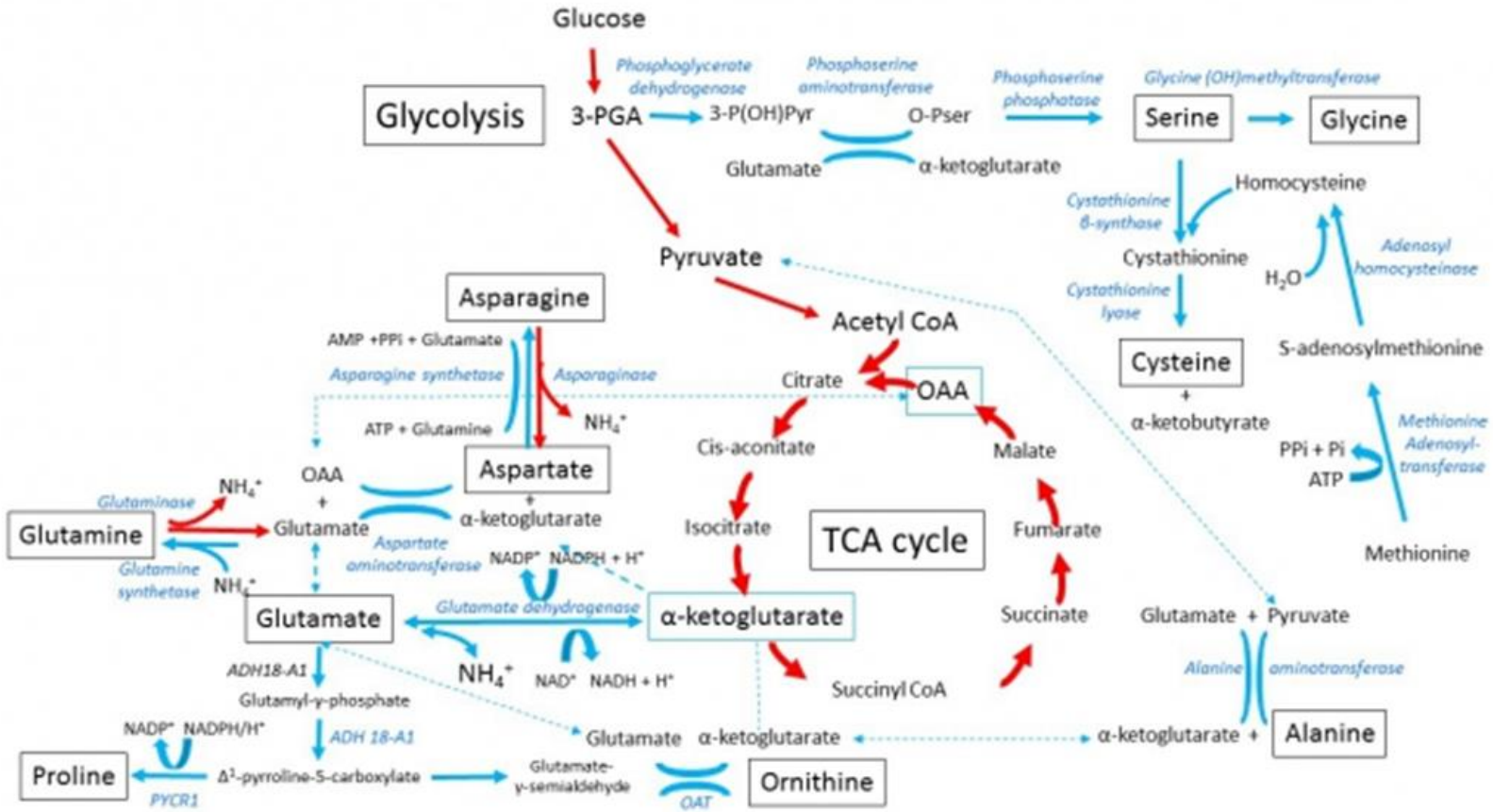
Hcy	Met	MMA	Diagnostic orientation
↗	N or ↘	↗	Vitamin B12 deficiency (dietary intake, malabsorption, N2O exposure); combined hereditary deficiency
↗	N or ↘	N	Folate deficiency; MTHFR deficiency; isolated remethylation defects
↗↗	↗	N	Cystathionine β-synthase deficiency
N	N	↗	Isolated methylmalonic aciduria; renal failure

Amino acid profiling



Accurate quantification requires **one labeled internal standard for each amino acid**.

Amino acid profiling



-Vue « large » du métabolisme
-Interprétations délicates
-Facteurs propres au patient
-Analyses de confirmations (OCT, Leucinoses …)

2. DIFFICULTIES

Standardization at every level: patient, analytical, interpretative

1

PATIENT

Pre-analytical

- Fasting, time of draw, diet
- Treatments, supplements, substances
- Tube, processing delay, temperature
- Aliquoting, freezing, storage

2

ANALYTICAL

Measurement

- Labeled internal standards, calibration
- Pooled QC samples throughout the run
- Randomized injection order across groups
- Batch effect monitoring and correction

3

INTERPRETATIVE

Data and reporting

- Normalization, multiple-testing control
- Annotation confidence
- Reference intervals, clinical context
- Transparent reporting of methods

> Most laboratory errors arise before the analysis itself

2. DIFFICULTIES

Patient-related factors to control or document



Fasting
8-12 h, water allowed

Time of day
Morning draw, at a fixed time

Physical activity
No hard exercise 24-48 h before

Posture and stress
Rest first, same position

Patients

Diet
Coffee, alcohol, last meal time

Treatments
Drugs, supplements (B9, B12…)

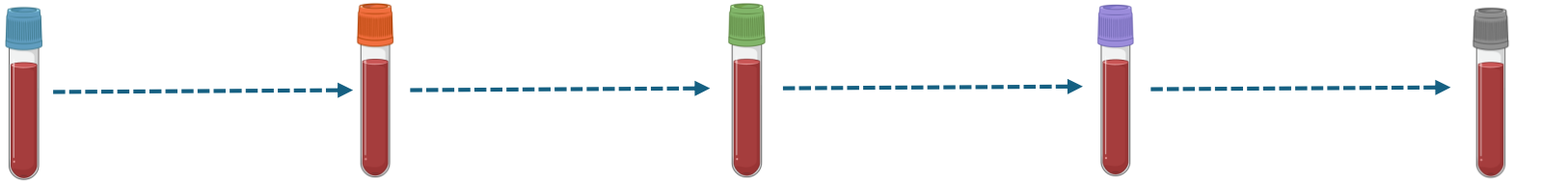
Tobacco and substances
Recent use, time since exposure

Posture and stress
Age, sex, BMI, menstrual cycle

Record every factor to adjust the statistical analyses

PRE-ANALYTICAL

Standardizing blood collection



Before

No gel separator tubes
They leach polymers that interfere with LC-MS

During

Tourniquet < 1 min
Adequate needle gauge, less hemolysis
Fill to the line
The blood-to-additive ratio sets the final concentration

After

Gentle inversions
As per the manufacturer, never shaking
Note the draw time
It starts the pre-centrifugation clock

1. PRE-ANALYTICAL

Serum and plasma are not interchangeable

PLASMA

Anticoagulated blood, can be centrifuged right away.

Less variability: no clotting step.

Contains an additive, a possible source of spurious signals.

Matrix of choice for multicenter studies.

SERUM

Clotting for 30 to 60 min at room temperature.

Platelet activation and cell metabolism during that time.

Often higher concentrations, a profile distinct from plasma.

Greater variability if clotting time is not fixed.

> One matrix and one tube type for the whole study, never mixed between groups

Every additive leaves its signature



EDTA

Chelates Ca^{2+} and Mg^{2+}

+ standard for untargeted LC-MS
+ inhibits metal-dependent enzymes
- Spurious NMR peaks, adducts in MS



Lithium heparin

Potentiates antithrombin

+ little NMR interference, reference for amino acids
- Possible ion suppression in LC-MS
- Inhibits PCR: a problem for multi-omics



Sodium citrate

Chelates Ca^{2+}, liquid additive

- 1:9 dilution: concentration bias
- masks endogenous citrate (Krebs cycle)

→ Keep for coagulation testing



Fluoride

Inhibits enolase (glycolysis)

+ stabilizes glucose, lactate, pyruvate
- Delayed action: glycolysis continues at first
- Promotes hemolysis: targeted use only

Choosing the tube for the analysis

Analysis	EDTA	Heparine	Citrate	Fluoride	Serum
Untargeted LC-MS	Recommended	Acceptable	Avoid	Avoid	Acceptable
Lipidomics	Recommended	Acceptable	Avoid	Avoid	Acceptable
1H NMR	Acceptable	Recommended	Avoid	Acceptable	Recommended
Glucose, lactate, pyruvate	On ice + < 30 min	On ice + < 30 min	Avoid	Recommended	Avoid
Total homocysteine	Recommended	Acceptable	Avoid	Acceptable	Avoid
Amino acids	Acceptable	Recommended	Avoid	Rarely used	Avoid

Choosing the tube for the analysis

Metabolite	Change at room temperature	Mechanism
Glucose	↘	Blood cell glycolysis
Lactate, pyruvate	↗	Blood cell glycolysis
Hypoxanthine, xanthine	↗	Purine catabolism
Arginine / ornithine	↘ Arginine, ↗ ornithine	Red-cell arginase
Lysophospholipids	↗	Export by red cells
Homocysteine	↗ About 10% per hour	LCAT activity, cell release

< 1 h
Centrifuge as soon as possible, ideally within 30 min

4°C
Tubes on ice between draw and centrifugation

Consistent
Same delay for every sample: consistency beats speed

PRE-ANALYTICAL

Freezing the sample and keeping it stable

- 80 °C

Long-term storage, or liquid nitrogen; avoid -20 °C

2-3 cycles

Of freeze-thaw per aliquot, ideally

< 2 h

From draw to freezing, with a standardized delay

Containers: screw-cap polypropylene cryotubes, freezer-proof labels

Shipping: dry ice and a temperature logger

Storage time: some classes drift over years; balance it across groups

Aliquots: small single-use volumes, so nothing is refrozen

Thawing: on ice, prompt analysis, autosampler at 4 °C

Light: protect light-sensitive compounds (folates, riboflavin, retinoids)

But will depend of each metabolites …

DETECTION

Two platforms: NMR spectroscopy and mass spectrometry.

Criterion	NMR	MASS SPECTROMETRY
Sensitivity	Limited (on the order of µmol/L)	High (nmol/L or lower)
Coverage	A few dozen major metabolites	Hundreds to thousands of metabolites
Reproducibility	Excellent; non-destructive technique	Good; matrix effects need to be controlled
Sample preparation	Minimal	Extraction, with prior chromatography
Quantification	Direct, using a single reference compound	Internal standards, ideally isotopically labeled
Typical use	Profiling of large cohorts	Targeted and untargeted analyses; clinical routine

TECHNIQUES

Principle of mass spectrometry





Separation
Liquid chromatography (LC) or gas chromatography (GC)

→

Source
Ionization: ESI, APCI (LC), electron impact (GC)

→

Analyzer
Sorts ions according to their m/z ratio

→

Detector
Measures ion abundance: mass spectrum

m/z

Mass-to-charge ratio, the quantity measured by the analyzer

Resolution

Ability to separate two ions with very close m/z values

Mass accuracy

Difference between measured and theoretical m/z, in ppm; points toward a molecular formula

Sensitivity

Lowest detectable amount; determines the limits of quantification

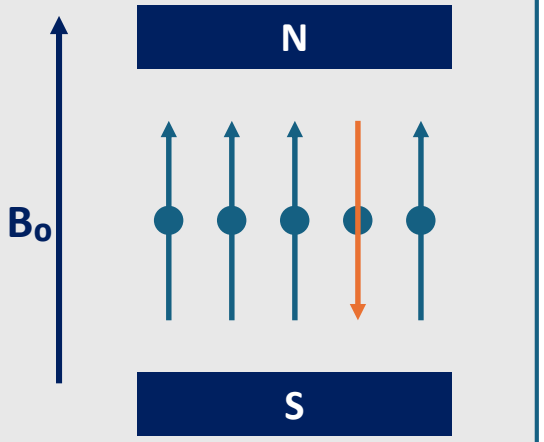
Triple quadrupole, QTRAP and QTOF

Criterion	Triple quadrupole	QTRAP	QTOF
Analyzers	Q1 – q2 – Q3	Q1 – q2 – Q3 or linear ion trap	Q1 – q2 – time-of-flight
Resolution	Unit, nominal mass	Unit, nominal mass	High, accurate mass (a few ppm)
Main mode	MRM	MRM + trap scans (EPI, MS^3)	Full scan, MS/MS (DDA, DIA)
Strengths	Sensitivity, robustness, quantification	Quantifies and confirms in the same run	Identification, retrospective data reanalysis
Typical use	Targeted, routine	Targeted with structural confirmation	Untargeted, research

TECHNIQUES

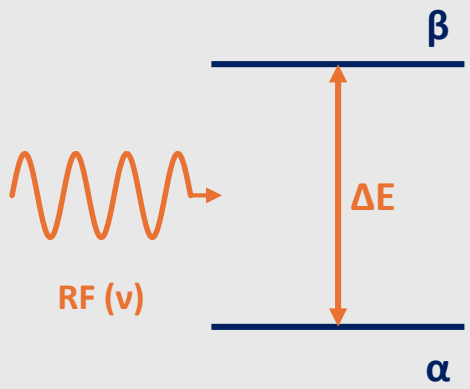
Principle of NMR spectroscopy

1 Magnetic field



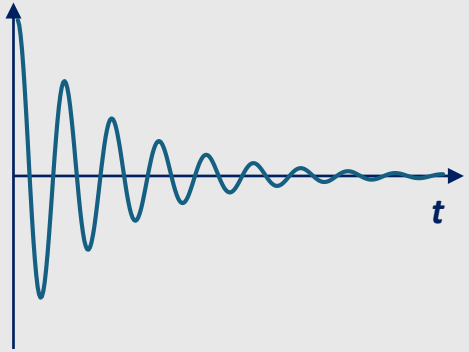
^{1}H nuclei align with B_0

2 RF pulse



Resonance when hν = ΔE

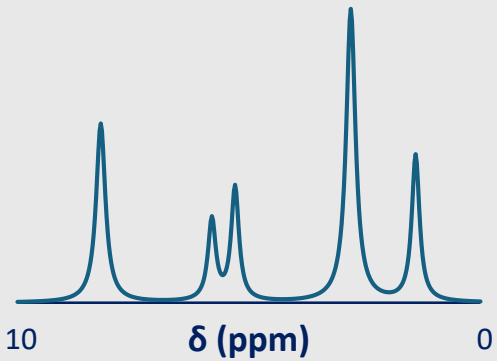
3 Relaxation



Decaying signal: the FID

FT

4 Fourier transform



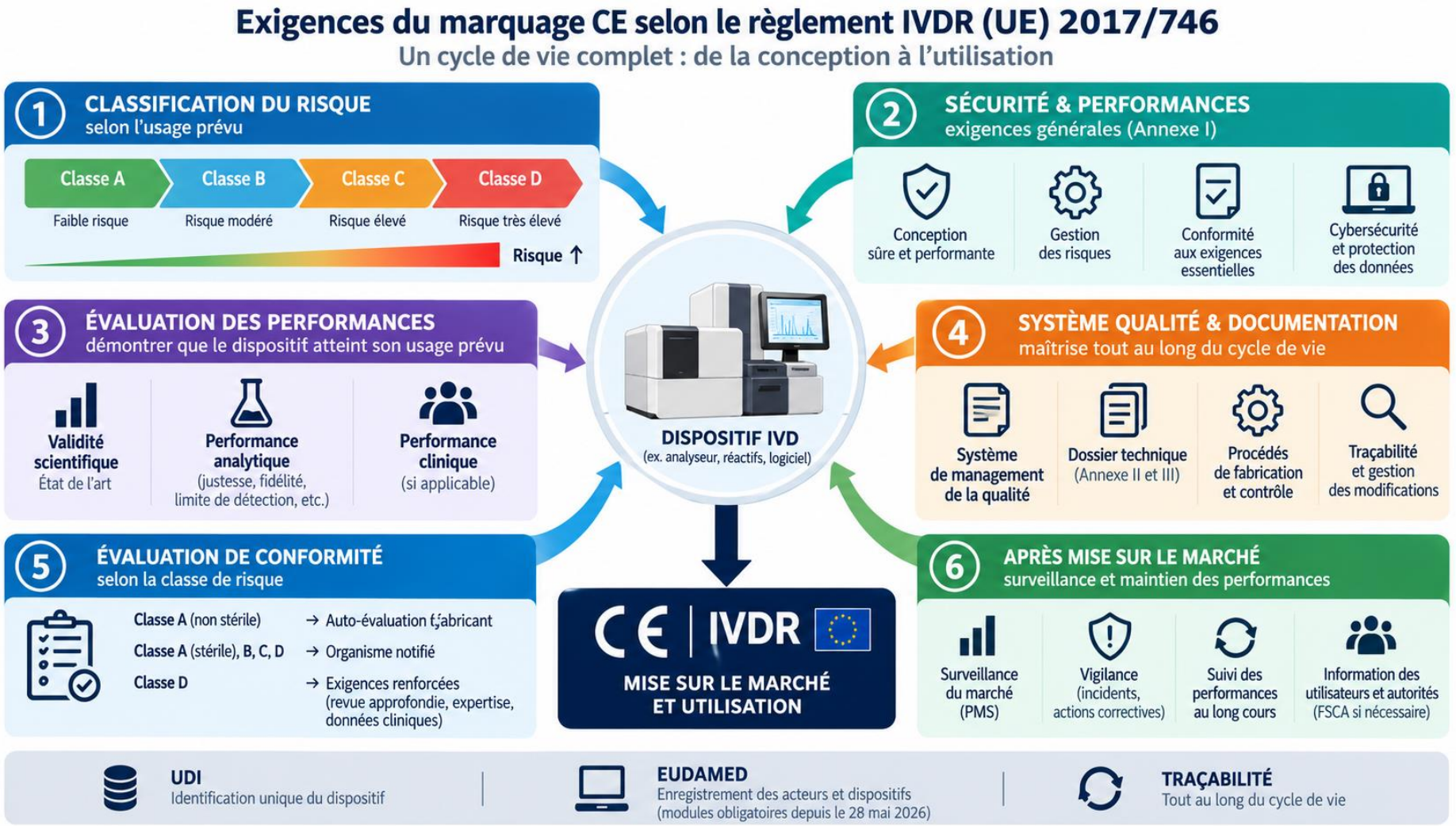
One peak per chemical environment

δ (ppm) → identification

Peak area ∝ concentration

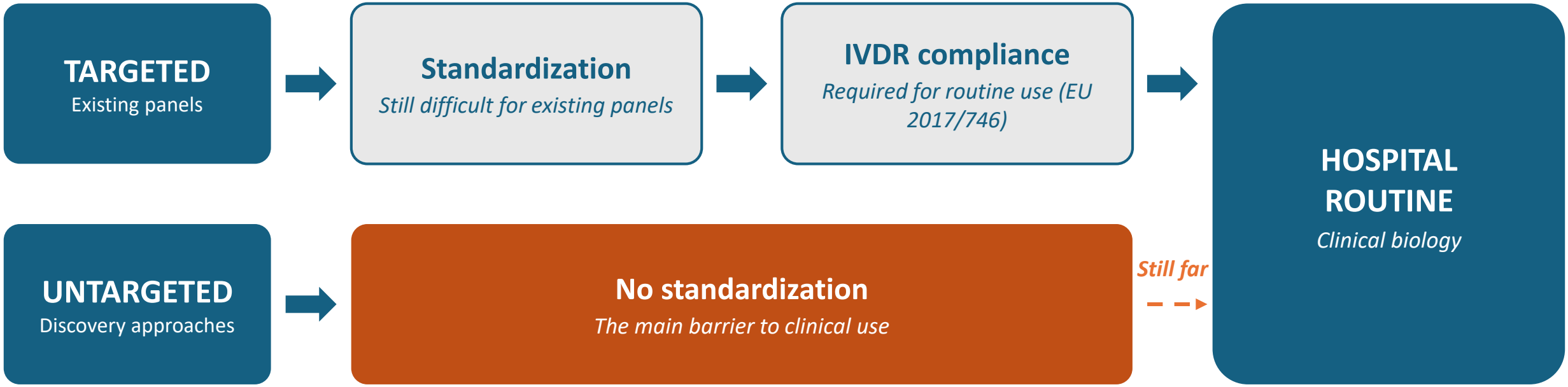
Non-destructive · minimal preparation

TECHNIQUES



+ marquages, suivi et habilitation de tout le matériel (enceintes, centri, piepettes …)

CONCLUSION



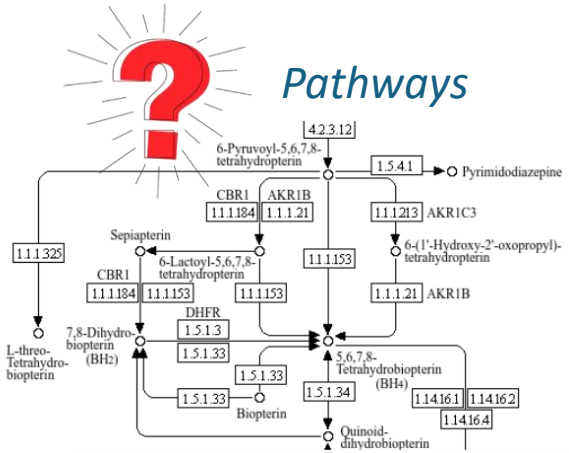
Targeted approaches must be mastered first. Untargeted metabolomics remains far from hospital biology, for lack of standardization.

PERSPECTIVE

TOP-DOWN
Untargeted → targeted



Untargeted
Broad profiling





Targeted
Confirmation, quantification

BOTTOM-UP
Targeted → untargeted



Targeted
Know panel

Nothing found
Unexplained phenotype



Untargeted
Open exploration

Discovery and validation feed each other: both strategies need the same standardization.